
Big Data

Management and Analytics

TU Clausthal, Institut für Informatik

Summer Semester 2026

Due Date: 13 May 2026, 16:00

Instructions — Please Read Carefully

In this assignment, we will use Neo4j again and practise role management and database security. For the exercises, we will use the “E-Commerce”-DB, which was already used in the fourth assignment and can be found on Moodle.

Cypher Reference: <https://neo4j.com/docs/cypher-refcard/current/>

Include all essential ideas and steps necessary to derive a solution. Write your solution in an understandable form or draw the structure with an appropriate tool. If there is missing information, make an assumption and state it explicitly — it will hold for the entire assignment unless declared otherwise.

Assignment Rules

- The assignment has a total of **10 marks**
- Work in your assigned teams of **up to 2 members**
- Submit your solutions **via Moodle**
- Solutions submitted after the deadline will automatically receive **0 marks**
- Copying solutions from or sharing with other students will lead to **exclusion from the course**
- The use of language software is **not permitted**; each group member must understand every part of the solution
- Document your solutions with **screenshots** and attach a **TEXT file** with your code/queries
- If anything is unclear, state your assumptions explicitly to make your solution understandable

Task 1 **5 Marks**

This task focuses on practising the **MapReduce** paradigm in Python, using the **map**, **filter**, and **reduce** operations. **Further Reading:**

- ▷ [Map, Filter, Reduce - learnpython.org](https://learnpython.org)
- ▷ [Regular Expressions \(re\) - Python Documentation](#)

(a) Create a Python class named **WordCounter**. In its constructor, read the contents of **dataset.txt** (passed when instantiating the class) and store them in a class variable. Implement separate methods for each of the following preprocessing operations:

- ▷ URL and email removal
- ▷ Conversion to lowercase
- ▷ Number removal
- ▷ Punctuation removal

Each method should process the stored data without modifying the original content. The output of every method must be stored separately and remain accessible even after all methods have been executed.

- (b) Implement a method that creates a **pipeline** of the preprocessing functions described above, where the output of each method serves as the input to the next. The output of this method will be used in Task 1c.
- (c) Add a method that implements the **mapping phase** of the MapReduce algorithm. Only map words that are more than two characters in length (all non-printable characters should be discarded). Refer to the lecture slides for the expected behaviour of the mapping phase.
- (d) Add a method that implements the **reduce phase** of the MapReduce algorithm. Refer to the lecture slides for the expected behaviour of the reduce phase.
- (e) Create an instance of the **WordCounter** class and execute each step, ensuring that the output of each step is saved in a separate **.txt** file. The final result of the reduce phase should be stored in a **.csv** file.

Note: All implementations must use only **built-in** Python operations and methods. The only permitted external package is **re**.

Deliverables: Six **.txt** files containing the results of each method, along with one **.csv** file containing the final output from the reduce phase. Results should be written to file using python code and not copy-paste.

Task 2 5 Marks

This task focuses on querying the **E-Commerce** database in Neo4j using **Cypher**. Write one Cypher query for each subtask and briefly explain what your query does.

- (a) Write a Cypher query that retrieves the first **5 products ordered by each user**. Return the username and the product name. Then modify the query so that it returns only the first **5 products ordered by name** that belong to the category “**Electronics**”.
- (b) Write a Cypher query to count how many products each user has purchased. Return the user ID and the purchase count, ordered from highest to lowest purchase count.
- (c) Write a Cypher query to find all reviews written by users. Return only the username, rating, and product name. Order the result by rating from highest to lowest.
- (d) Write a Cypher query to find all users who added products to their wishlist but never purchased those same products. Return only users who have more than **4** such products, together with the list of wishlisted-but-never-purchased products.

Deliverables: Submit a `.txt` file containing all Cypher queries and screenshots showing the execution results in Neo4j.

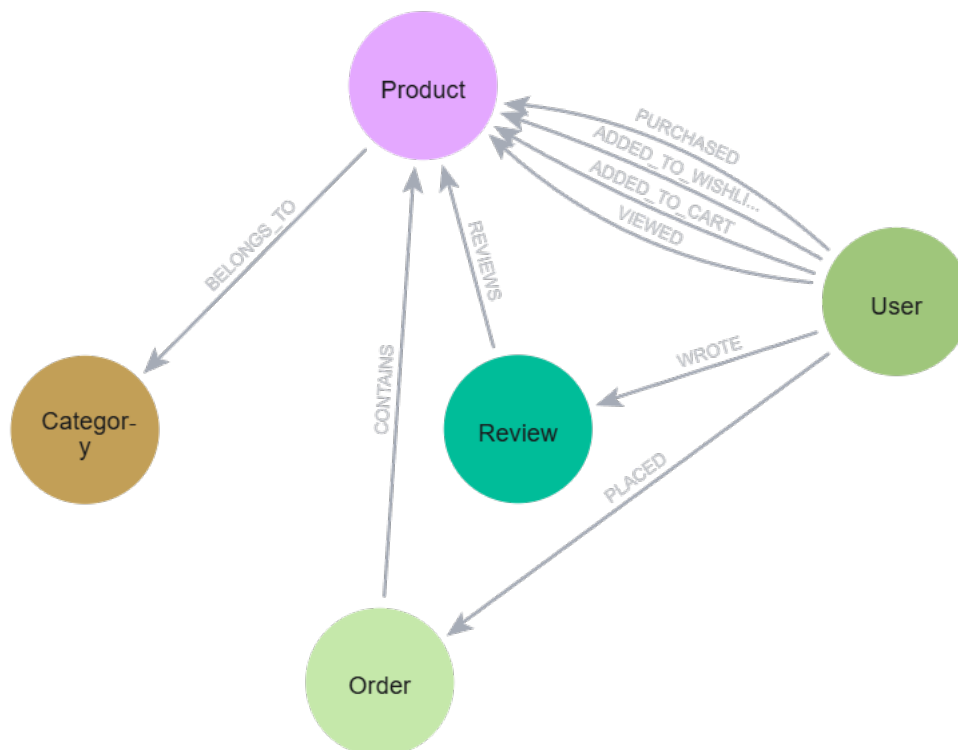


Figure 1: E-commerce schema